What's the point?

Steven Breger



"Believe in the universe." –Mr. Scheiman

1 Introduction

Mathematics is infamously difficult. It starts off with simple counting and operations, and quickly begins to blow up. Often, even the most simple looking of questions require very long, difficult, and non-elementary solutions. We don't need to look further than the counting numbers (the things that kindergartners know about!): simple-looking questions like the Collatz conjecture or the twin prime conjecture are still not solved, despite being explainable to a well-versed elementary schooler.

Why then, you may ask, is it even worth studying mathematics? Doesn't it feel hopeless and pointless? Besides, most of us won't even need something as simple as high-school algebra throughout our lives, let alone basic calculus, rigorous analysis, or the countless branches of essentially inapplicable pure math. Even if you want to be an engineer, requiring an understanding of algebra, calculus, and physics, it seems pointless to go deeper into pure math. After all, when would an engineer need to explore the ultra theoretical spaces studied within topology or know what the Riemann hypothesis implies about prime asymptotics? Shouldn't we just leave all of this crazy stuff to the few hyper-geniuses of the world that want to lock themselves in a room all day and learn all this theory for it's own sake?

Well, according to Mr. Scheiman, who teaches calculus and differential equations at Stuyvesant, doing so would be a grave mistake. Seeing how Mr. Scheiman's teaching journey has influenced his perspective on mathematics, we can get a sense of why he does math, and why we all should engage with it too.

2 Connection

Besides his teaching at Stuyvesant, Mr. Scheiman simultaneously teaches at Rutgers University, where he earned his mathematics degree. As you can imagine, teaching at Stuy until around 3, commuting for an hour, and then teaching afternoon classes at the university. When asked why he does this, Mr. Scheiman said "Of course I need to pay the rent, but I've always loved it. I think it's about being in front of a group of people in a lecture and helping them out... I get an excitement from it." Mr Scheiman added that he especially loves the daily tight-knit environment in the Stuy high-school classrooms: "Here I can say a witty joke, something sarcastic, and they will understand. You might not get that in college where there's large lecture halls and students are taking classes just because they need to, unlike here where students are taking math classes as electives and there's a general liking to it."

Surprisingly, human connection is one of the biggest benefits of doing math. We stereotypically see math as a solitary art, but it doesn't have to be. As Mr. Scheiman points out, a big part of teaching it is that human connection with your students. Importantly though, you don't even need to be a teacher or have a student to connect over math. Anyone can do it. Working on a problem set with your peers, talking about a problem with someone, or explaining something to someone who is stuck, is a shockingly effective way to meet new people and create new friendships. In fact, one of Mr. Scheiman's favorite ways to bond with his students is to make jokes about "difficult–I mean differential equations" or to start off class by saying "GooooOOOOOOOOOOOOD afternoon everyone!" He points out that it's important to know the students, read the energy level in the room, and adjust to create the best relationship possible with the people you do mathematics with. That way, both you and they can get the most out of it.

3 Thinking is Power

Aside from the fact that math lets you make so many new friends, it improves your problem solving and ability to understand complex ideas. Whether you are training for high-school math competitions, explaining algebra to your brother, or reading a topology textbook, there is some part of you that is thinking, gaining an intuition for approaching difficult problems, and learning how to explain what you see. Mr. Scheiman noted how these problem-solving skills are always useful, saying that "Math is the great equalizer. People won't take advantage of you. Car salesmen, accountants, things like that. The more math you know, the better you are in the world. I say that first and foremost."

These abilities of critical thinking are useful regardless of what you do. Sadly, this viewpoint is not commonly expressed because people often think that math is just about calculation, about solving some crazy long equation with a set procedure that was taught to you by someone else. But in reality, math is much more like an art form. Calculation is merely a means of obtaining data, intuition, and deductions in order to find a new proof, make a discovery, and think more deeply about a problem. If you view math in this way it, the skills you gain from math quickly become applicable to a wide range of disciplines, even if the exact math itself isn't. Don't believe it? See for yourself:

- If you read literature, learning about logic, mathematical proof, and deduction can help you draw more interesting conclusions, perhaps giving you the capacity to connect and analyze more pieces of evidence at once.
- If you are a construction worker, well each construction site is essentially its own giant problem. There's a lot of problem solving that goes into what parts to use where, how to adjust something without tearing down an existing wall, and the best way to budget. It's the same kind of thinking that goes on in a hard math problem.
- If you are a chef there's countless times when you have to figure out the best way to create a certain flavor, texture, or style. Going through all the possibilities and having an intuition for what works is the same skill that used in many math problems.

The list goes on. Mr. Scheiman put it best when he said that "if the world is more math literate, the world would be a better place."

4 Conclusion

So, to answer the question, the point is that no one, and I really do mean no one, can ever know enough math. Learning more math is always a great thing to do so long as you are a living breathing human. Math satisfies natural instincts of curiosity, creates a surprising amount of socialization, and fosters critical thinking, useful in any part of life. Yes, it's useful even if you don't need to solve daily difficult–I mean differential equations.